OSRCORE

Robotics Development Board

CE FC RoHS

For robotics and embedded control development

Designed for mobile robots, education, research and motion control, with an ESP32-S3 controller, inertial and magnetic sensors, and peripheral interfaces.

An integrated USB hub, 100 Mbps Ethernet and CAN bus support host connectivity, sensor acquisition and actuator control.

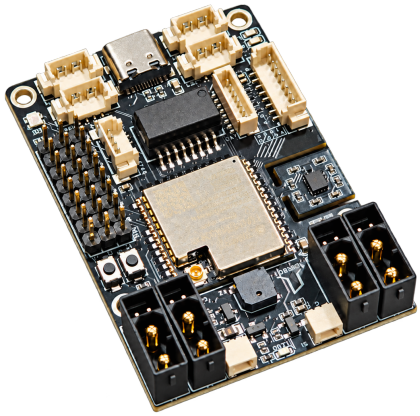


Figure 1-1 Product appearance

ESP32-S3	9–26 V	9 axes	USB HUB
Controller	Wide input range	Inertial & magnetic	Peripheral ports

GPIO expansion Multiple GPIO connections support peripherals and custom functions.

Onboard sensing Acquire three-axis acceleration, angular velocity and magnetic field data.

Integrated connectivity USB, Ethernet, CAN and expansion interfaces simplify robot integration.

Applications

Illustrative applications



Quadruped robots

Attitude sensing and peripherals



Biped robots

Multi-joint control development



Robotic arms

Actuator and sensor expansion



Unmanned vehicles

Chassis control and connectivity

1 Hardware configuration

The controller, sensors and peripheral interfaces combine low-level control and state acquisition on one board.

Controller and sensors

Item	Specification
Module	ESP32-S3-WROOM-1U-N16R8
Processor	Dual-core Xtensa LX7
Clock	240 MHz
Flash / PSRAM	16 MB / 8 MB
6-axis IMU	QMI8658A
3-axis mag.	QMC6309
Wireless	2.4 GHz
Antenna	External antenna

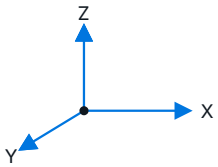
Electrical and connectivity

Item	Specification
Power input	9–26 V DC
USB host	USB Type-C / MX1.25 4P
USB ports	3 ports · MX1.25 4P
Wired links	100 Mbps Ethernet / CAN
GPIO	EX1 / EX2 / EX3 / EX4
Dimensions	60 × 41 mm
Weight	21 g
Temperature	0–55 °C

QMI8658A

Six-axis inertial sensing

Three-axis acceleration and angular velocity provide inertial data for robot state estimation.

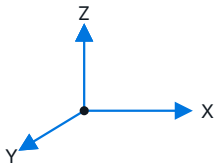


Acceleration X/Y/Z · Angular velocity X/Y/Z

QMC6309

Three-axis magnetic sensing

Measures the magnetic field along three axes for use as an observation input in heading estimation.



Magnetic field X/Y/Z

2 Interface overview

Use the numbered photographs to locate connectors and onboard components. See Pin definitions for wiring details.

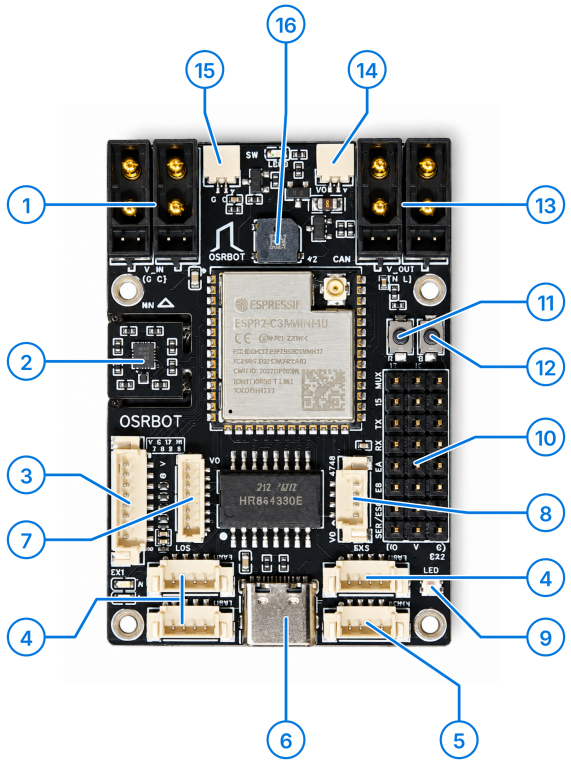


Figure 2-1 Front connectors and components

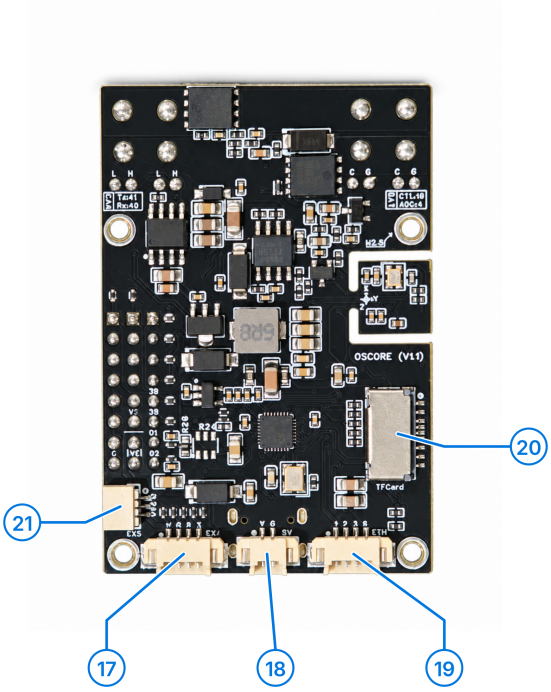


Figure 2-2 Rear connectors and components

100 Mbps Ethernet: LDS (7) and ETH (19) share one 100 Mbps Ethernet link. Use either connector. LDS is the board silkscreen label.

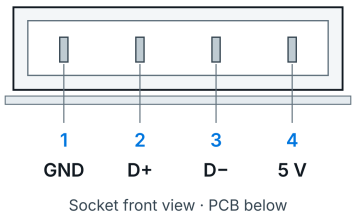
No.	Name	Label / specification	Function
1	Power input	V_IN · XT30 2+2 × 2	9–26 V DC input with switch control contacts
2	Inertial and magnetic sensors	QMI8658A / QMC6309	6-axis inertial and 3-axis magnetic sensing
3	EX1 expansion	MX1.25 8P	6 GPIO signals and 5 V power
4	USB peripheral ports	USB1 / USB2 / USB3 · MX1.25 4P	3 downstream USB ports
5	USB host	USBIN · MX1.25 4P	Upstream USB; use either this port or USB-C
6	USB Type-C	USB-C	Shares the upstream data link with USBIN

No.	Name	Label / specification	Function
7	100 Mbps Ethernet	LDS · SH1.0 8P	100 Mbps Ethernet with main power output
8	EX3 expansion	MX1.25 4P	GPIO 47 / GPIO 48
9	RGB indicator	WS2812	Programmable RGB LED
10	EX2 expansion header	2.54 mm · 3 × 8P	10 GPIO signals and power connections
11	Reset button	R	Resets the ESP32-S3
12	BOOT button	B	Connected to GPIO 0
13	Power and CAN output	V_OUT · XT30 2+2 × 2	Main power output and CAN H / L
14	Auxiliary power output	VO · SH1.0 2P	Main power output through a resettable fuse
15	External switch	SW	External power switch control
16	Passive buzzer	GPIO 42	PWM-driven audible indication
17	EX4 expansion	MX1.25 4P	GPIO 3 / GPIO 9
18	5 V output	5V · MX1.25 2P	5 V peripheral supply
19	100 Mbps Ethernet	ETH · MX1.25 4P	Shares the Ethernet link with LDS
20	MicroSD slot	TFCard	Memory card access through the USB host
21	EX5 LED expansion	3P	Cascade output from the onboard WS2812

3 Pin definitions

Socket drawings show the board-side opening with the PCB below. Numbers identify pins; signals read from left to right. EX2 and XT30 2+2 have separate viewing directions.

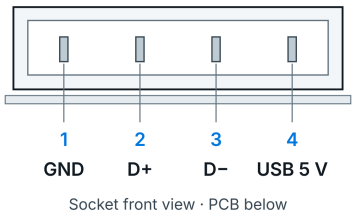
USB1 / USB2 / USB3 · MX1.25 · 4P



Socket front view · PCB below

Pin	Signal
1	GND
2	USB D+
3	USB D−
4	5 V

USBIN · MX1.25 · 4P

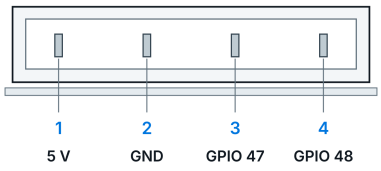


Socket front view · PCB below

Shares the upstream data lines with USB-C.

Pin	Signal
1	GND
2	USB D+
3	USB D−
4	USB 5 V

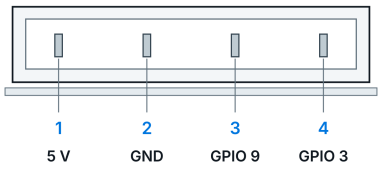
EX3 · MX1.25 · 4P



Socket front view · PCB below

Pin	Signal
1	5 V
2	GND
3	GPIO 47
4	GPIO 48

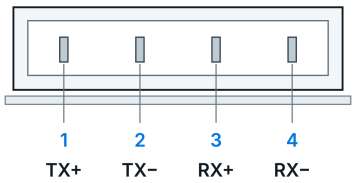
EX4 · MX1.25 · 4P



Socket front view · PCB below

Pin	Signal
1	5 V
2	GND
3	GPIO 9
4	GPIO 3

Ethernet · ETH · MX1.25 · 4P

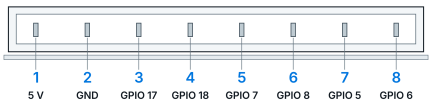


Socket front view · PCB below

Shares one Ethernet link with LDS. Use either connector.

Pin	Signal
1	ETH TX+
2	ETH TX−
3	ETH RX+
4	ETH RX−

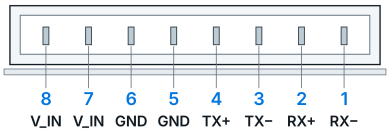
EX1 · MX1.25 · 8P



Socket front view · PCB below

Pin	Signal
1	5 V
2	GND
3	GPIO 17
4	GPIO 18
5	GPIO 7
6	GPIO 8
7	GPIO 5
8	GPIO 6

Ethernet · LDS · SH1.0 · 8P

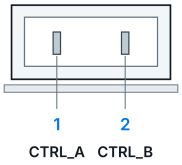


Socket front view · PCB below

Pins 7 and 8 carry the main input voltage (9–26 V).

Pin	Signal
1	ETH RX−
2	ETH RX+
3	ETH TX−
4	ETH TX+
5	GND
6	GND
7	V_IN
8	V_IN

External switch · SW · SH1.0 · 2P

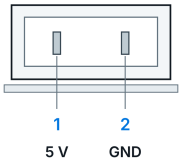


Socket front view · PCB below

Pin	Signal
1	CTRL_A
2	CTRL_B

Connect an external power switch.

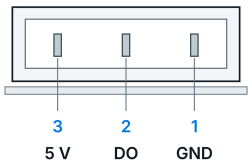
5 V power output · MX1.25 · 2P



Socket front view · PCB below

Pin	Signal
1	5 V
2	GND

LED expansion · EX5 · 3P

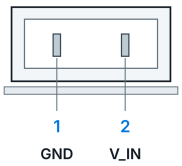


Socket front view · PCB below

Pin	Signal
1	GND
2	WS2812 DO
3	5 V

DO is the cascade output from the onboard WS2812.

Auxiliary power · VO · SH1.0 · 2P



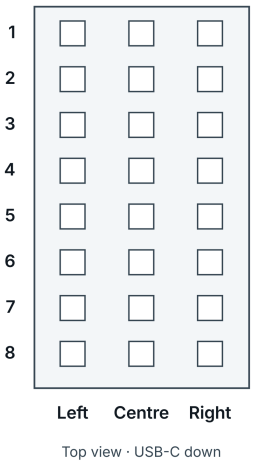
Socket front view · PCB below

Pin	Signal
1	GND
2	V_IN

The output follows the main input voltage (9–26 V).

3.1 EX2 · 2.54 mm header

View from the front with USB-C facing down. Rows 1–8 run from top to bottom.



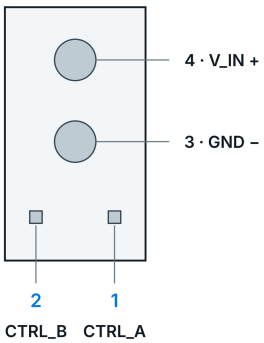
Top view · USB-C down

Row	Left	Centre	Right
1	GPIO 13	GPIO 12	GPIO 14
2	GPIO 15	5 V	GND
3	GPIO 43	5 V	GND
4	GPIO 44	5 V	GND
5	GPIO 39	5 V	GND
6	GPIO 38	5 V	GND
7	GPIO 1	Separate supply	GND
8	GPIO 2	Separate supply	GND

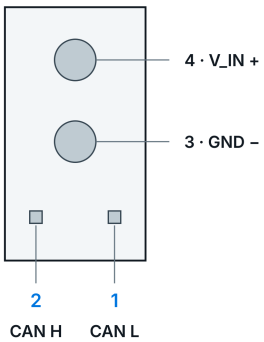
All three pins in row 1 are GPIOs. The centre pins in rows 7 and 8 form a separate supply connection, distinct from the 5 V column.

3.2 Power and CAN

V_IN · Main power input



V_OUT · Power and CAN output



Front view, power end at top. Both XT30 2+2 connectors in each pair share the pinout.

Interface	Pin	Signal	Function
V_IN	1 / 2	CTRL_A / CTRL_B	Switch control contacts
V_IN	3 / 4	GND_BAT / VCC_BAT	Main power negative / positive
V_OUT	1 / 2	CAN L / CAN H	CAN differential signals
V_OUT	3 / 4	GND / V_IN	Power negative / positive
VO	1 / 2	GND / V_IN	Auxiliary power output

VO uses SH1.0 2P; V_IN and V_OUT use XT30 2+2. GPIO logic level is 3.3 V.

3.3 Onboard function pins

Function	Signal	Controller pin
Onboard I²C	SDA / SCL	GPIO 10 / GPIO 11
IMU interrupt	INT	GPIO 21
CAN	TX / RX	GPIO 41 / GPIO 40
Onboard USB	D− / D+	GPIO 19 / GPIO 20
Passive buzzer	PWM	GPIO 42
RGB LED	WS2812 DIN	GPIO 45
Battery voltage sensing	ADC	GPIO 4
Power control	CTRL	GPIO 16
BOOT button	BOOT	GPIO 0
Reset button	RESET	CHIP_PU / EN

Expansion-connector GPIO assignments are listed earlier in this chapter.

4 Power and onboard functions

4.1 Power and switching

Main input: 9–26 V DC. V_OUT, LDS pins 7/8 and VO carry the input voltage. The onboard buck converter supplies 5 V to peripherals.

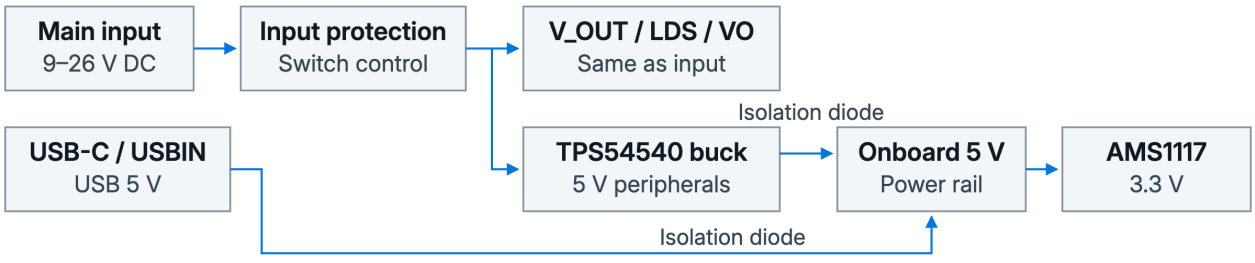


Figure 4-1 Power distribution

The input circuit provides reverse-polarity protection, soft start and overvoltage protection. Connect an external switch to SW; GPIO 16 connects the controller to the power control circuit.

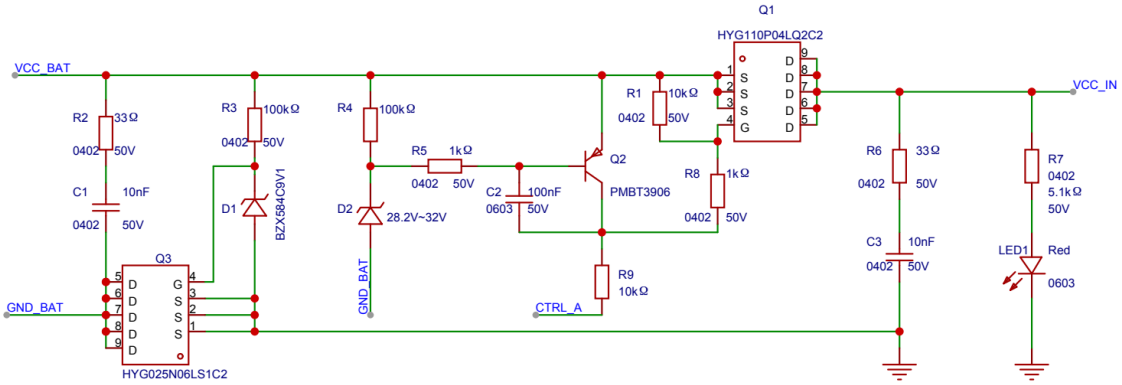


Figure 4-2 Input protection circuit

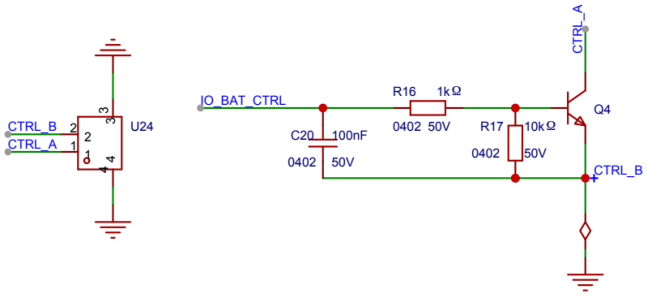


Figure 4-3 Power switch circuit

When powered only by USB, V_OUT, LDS, VO and the 5V peripheral power port cannot be used as power outputs.

4.2 USB networking and storage

USB-C and USBIN share one upstream interface. USB1, USB2 and USB3 connect peripherals. The onboard USB hub and bridge connect the ESP32-S3, 100 Mbps Ethernet and MicroSD to the USB host.

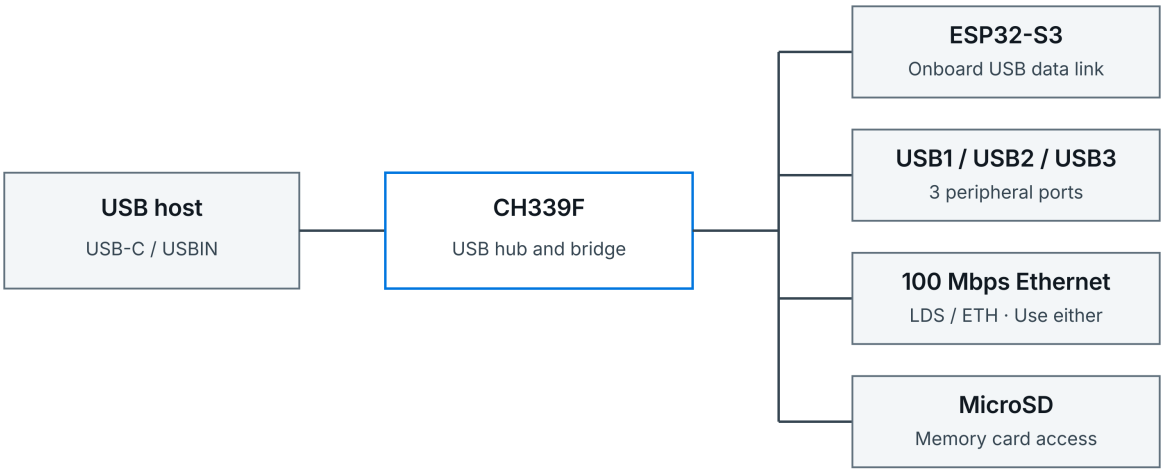


Figure 4-4 USB and peripheral connections

4.3 Sensors and CAN

The QMI8658A 6-axis inertial sensor and QMC6309 3-axis magnetometer share an I²C bus. SDA and SCL connect to GPIO 10 and GPIO 11 respectively.

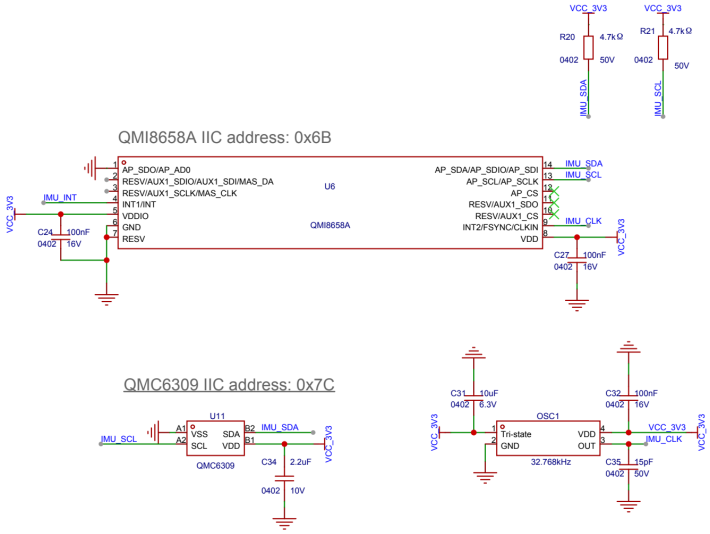


Figure 4-5 Nine-axis sensor circuits

The CAN transceiver connects to GPIO 41 (TX) and GPIO 40 (RX). CAN H and CAN L are available at V_OUT.

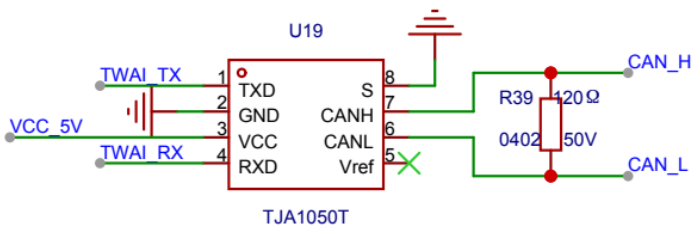


Figure 4-6 CAN transceiver circuit

4.4 Buttons indicators and buzzer

The R button pulls CHIP_PU low to reset the controller. The B button connects to GPIO 0 for boot-mode selection.



Figure 4-7 Reset and BOOT buttons

WS2812 RGB LED

GPIO 45 controls the onboard RGB LED. EX5 provides the cascade signal for additional LEDs.

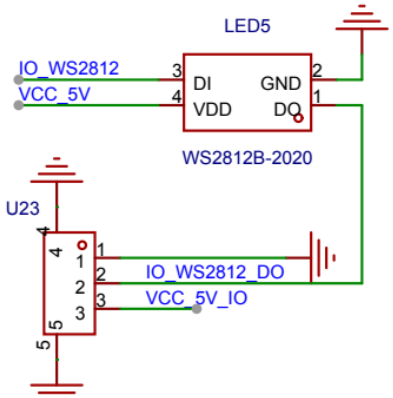


Figure 4-8 RGB LED and cascade interface

Passive buzzer

GPIO 42 drives the passive buzzer through a transistor using PWM.

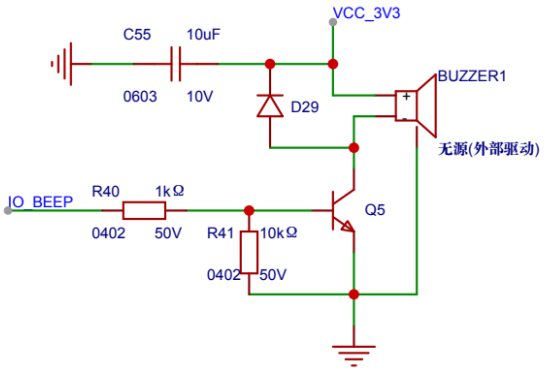


Figure 4-9 Buzzer driver circuit

Robot connection example



Figure 4-10 Robot connection example · USB / UART

5 Dimensions and installation

Allow clearance for rear components and wiring. Keep supports away from solder joints.

5.1 Installation and wiring

Mount the board Secure the board using its four mounting holes, leaving clearance for rear components, the MicroSD card and wiring.

Connect peripherals Disconnect power before wiring. Follow the pin definitions and distinguish main power, 5 V power and 3.3 V GPIO signals.

Check and power on Verify input polarity, connector orientation and peripheral supply requirements before applying 9–26 V main power.

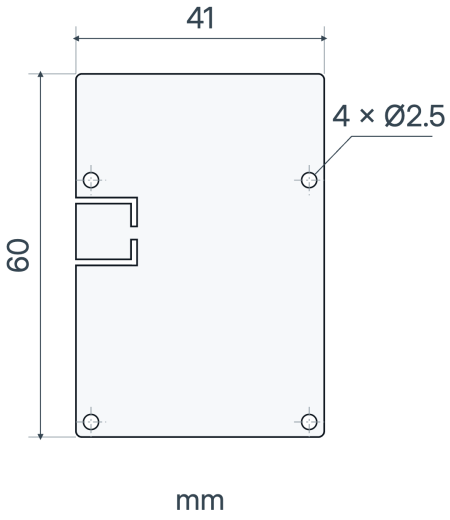


Figure 5-1 Dimensions and mounting holes

Mounting specifications

Item	Specification
Dimensions	60 × 41 mm
Mounting holes	4 × Ø2.5 mm
Weight	21 g
Temperature	0–55 °C

6 Development resources

OSRCORE example projects

Development examples and documentation.

Scan for example projects

github.com/osrbot/ Development-Board-OSRCORE-Example

OSRBOT · OSRCORE · Development Board

For wireless operation, keep the antenna at least 20 cm away from people.
Product configurations and specifications are subject to change without prior notice.